SET NO. 22

TIME SERIES ANALYSIS

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Course: DSA0610 Data Handling and Visualization

Dataset: Monthly Sales Data

Month	Sales ($)
January	15,000
February	18,000
March	22,000
April	20,000
May	23,000

Question 1: Time Series Line Chart

Aim:

To create a line chart to visualize the trend in monthly sales.

Algorithm:

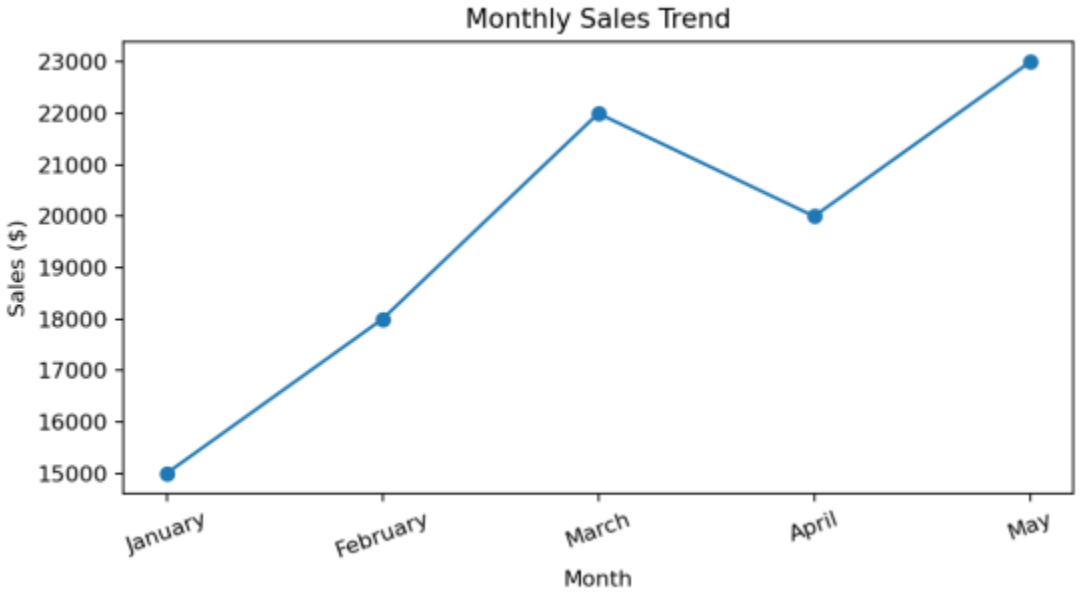
1. Store the month names and sales values.
2. Plot sales using a line chart.
3. Label the axes and add a title.
4. Display the chart.

Code (R):

```
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(sales, type="o", col="blue", xaxt="n",
    xlab="Month", ylab="Sales ($)",
    main="Monthly Sales Trend")
axis(1, at=1:5, labels=month)
```

Output: The line chart shows monthly sales from January to May.



Result: Sales generally increased from $15,000 in January to $23,000 in May, with a decrease in April.

Question 2: Advertising Budget and Sales Scatter Plot

Aim: To analyze the relationship between advertising budget and monthly sales using a scatter plot.

Algorithm:

1. Store advertising budget and sales values.

2. Plot advertising on the x-axis and sales on the y-axis.

3. Add axis labels and a title.

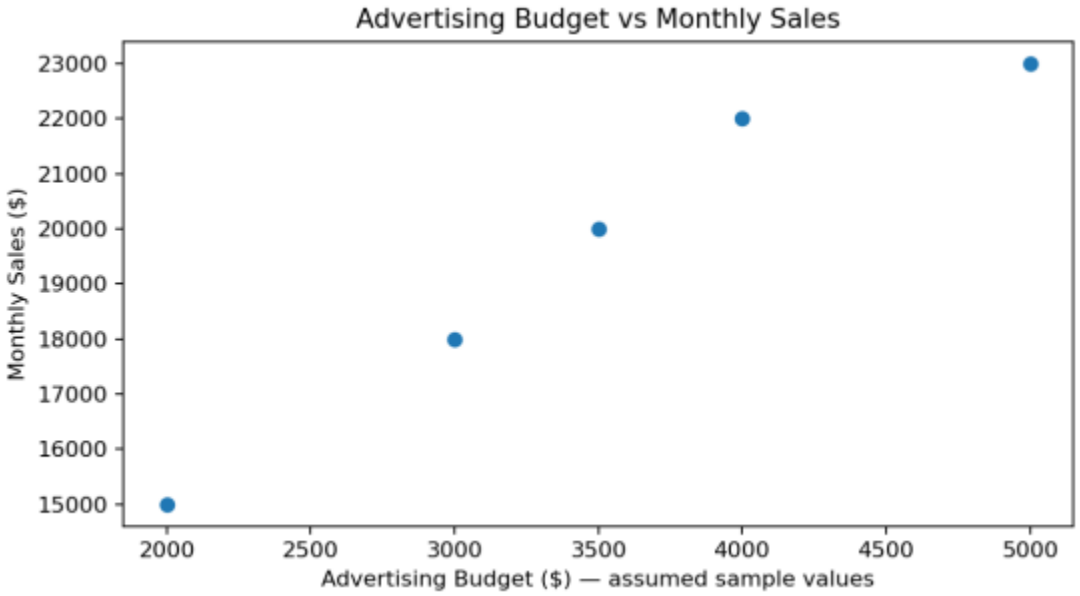
4. Interpret the pattern.

Code (R):

```
advertising <- c(2000, 3000, 4000, 3500, 5000)
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(advertising, sales,
    main="Advertising Budget vs Sales",
    xlab="Advertising Budget ($)",
    ylab="Monthly Sales ($)",
    pch=19, col="blue")
```

Output: A scatter plot showing advertising budget against monthly sales.



Result: The assumed data suggest a positive relationship: higher advertising budgets are generally associated with higher sales. This does not prove that advertising causes sales to increase.

Question 3: Tableau Sales Dashboard

Aim: To build a dashboard combining a monthly sales line chart and a pie chart showing product-wise sales distribution.

Algorithm:

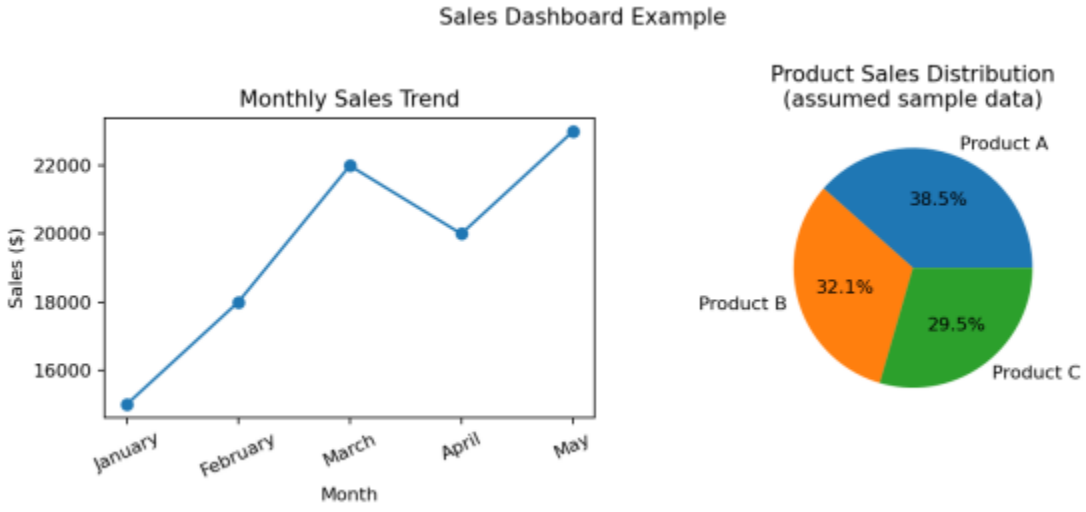
1. Connect the dataset in Tableau.

2. Create a line-chart worksheet by placing Month on Columns and Sales on Rows.

3. Create a pie-chart worksheet using Product for Color and Sales for Angle.

4. Add both worksheets to a dashboard. 5. Add a title and save.

Output: A dashboard example showing the monthly sales trend and product sales distribution.



Result: The dashboard combines time-based and product-based views. Replace the assumed product values with actual product data for a real analysis.

Question 4: Autocorrelation Plot

Aim: To create an autocorrelation function (ACF) plot to examine patterns in monthly sales.

Algorithm:

1. Store the monthly sales values.

2. Convert them into a time-series object.

3. Use acf() to calculate autocorrelation at different lags.

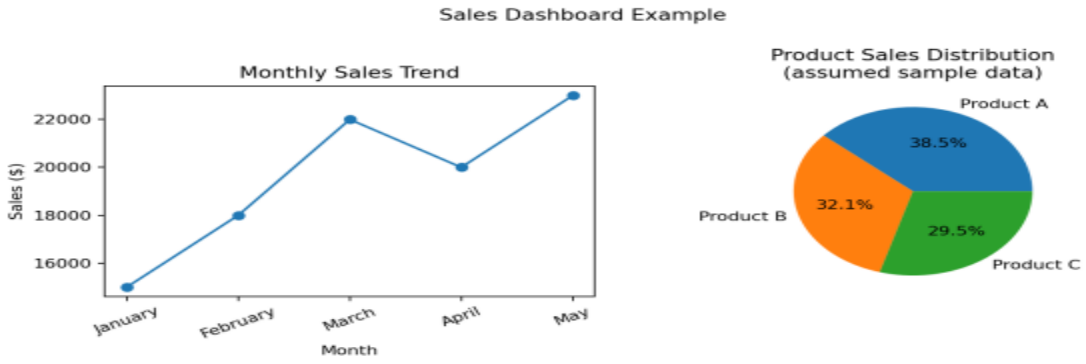
4. Observe the plot for repeating patterns.

Code (R):

```
sales <- c(15000, 18000, 22000, 20000, 23000)
sales_ts <- ts(sales, frequency=12)

acf(sales_ts, lag.max=3,
   main="Autocorrelation of Monthly Sales")
```

Output: An ACF plot displaying the correlation of the sales series with its lagged



Result: Only five months of data are available, so annual seasonality cannot be identified reliably. Several years of monthly data are recommended to confirm seasonal patterns.